Resume of *SAJAL HALDER, PhD*

Current Position
Research Fellow (B2)
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ResearchGate Link, Google Scholar Link

PROFESSIONAL OVERVIEW

Research Fellow and cybersecurity scientist with extensive experience in AI-driven systems security, software vulnerability analysis, and large-scale security analytics. Proven record of delivering high-impact cybersecurity research spanning AI-enabled security, critical software infrastructure, and vulnerability detection, with publications in top-tier security and computer science venues. Experienced in collaborative research with CSIRO Data61, government, and industry partners, including Google-funded national projects. Demonstrated capability in scientific leadership, PhD supervision, and translating research into deployable security solutions that strengthen national digital resilience.

KEY SKILLS

- **Cybersecurity & Systems Security:** Software Vulnerability Detection, Secure Systems Design, AI for Cybersecurity, Critical Infrastructure Security (Software Supply Chain, OSS, Dependency Networks), Security Analytics, Threat Detection, Trust & Safety Systems.
- **AI & Computational Modeling:** Deep Learning (PyTorch, TensorFlow), Natural Language Processing (NLP), Transformer Models, Large Language Models (LLMs), Advanced Predictive Modelling for Biomedical Applications, Graph-Based Learning, Multi-modal Data Integration.
- **Academic Leadership & Teaching:** Curriculum Development (Undergraduate & Postgraduate), Course Coordination, Online Learning Delivery, PhD & Higher Degree Research (HDR) Supervision, Student Mentorship, Pedagogical Improvement informed by Student Feedback.
- **Strategic Research & Engagement:** High-Impact Publications (IEEE, ACM, Elsevier, Springer), Competitive Research Funding (ICT Ministry, University Grants), Industry & Government Liaison (CSIRO Data61, Google, DSTG), Multidisciplinary Collaboration, Scientific Reporting.
- **Programming & Technical Platforms:** Python, PyTorch, TensorFlow, Hugging Face, BERT/DeBERTa, LLaMA-3.1, Gemini API, LangGraph, LangChain, MLOps/LLMOps (GCP, AWS).

SELECTED PUBLICATIONS

1. **Sajal Halder** M. Bewong, A. Mahboubi, Y. Jiang, R. Islam, MD Zahid Islam, R. I. H. Leung, E. Ahmed, G. Ramachandran, and A. Babar, Malicious package detection using metadata information, In The Web Conference 2024. **Core Rank: A*** , Publisher: ACM
2. **Sajal Halder**, Muhammad Ejaz Ahmed, Seyit Camtepe, FuncVul: An Effective Function Level Vulnerability Detection Model using LLM and Code Chunk. In the 30th European Symposium on Research in Computer Security (ESORICS) 2025, **Core Rank: A**.
3 Michael Robinson, **Sajal Halder**, Muhammad Ikram, Hyoungshick Kim, Seyit Camtepe, Muhammad Ejaz Ahmed, Original Sin of npm: A Study on Vulnerability Propagation in JavaScript Dependency Networks. In 21st ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2026), **Core Rank: A**.
4. Sishun Liu, **Sajal Halder**, Ke Deng, Yan Wang and Xiuzhen Zhang. Unsupervised Anomaly Detection of Information Operations Users via Behavioral and Language Patterns. In ECML-PKDD, 2026, **Core Rank: A**.

5. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang.Deep learning of dynamic POI generation and optimisation for itinerary recommendation. In **ACM Transactions on Recommendation Systems**, 2025.
6. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang.A Survey on Personalized Itinerary Recommendation: From Optimisation to Deep Learning. In Applied Soft Computing Journal, 111200, 2023, **Q1 Ranked Journal**, (Impact Factor – 8.7). Publisher: Elsevier.
7. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. Capacity-aware fair POI recommendation combining transformer neural networks and resource allocation policy. In Applied Soft Computing Journal, pages 110720, 2023, **Q1 Ranked Journal**, (Impact Factor – 8.7). Publisher: Elsevier.
8. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. POI Recommendation with Queuing Time and User Interest Awareness. In Data Mining and Knowledge Discovery, 2022, **Q1 Ranked Journal**, (Impact Factor - 5.406). Publisher: Springer.
9. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. Efficient Itinerary Recommendation via Personalised POI Selection and Pruning. In Knowledge and Information Systems, Volume 64, Pages 963–993, 2022. **Q1 Ranked Journal**, (Impact Factor - 2.53), Publisher: Springer.
10. **Sajal Halder**, Nahina Islam, Biplob Ray, Elizabeth Andrews, Pushpika Hettiarachchi, Emma Jackson. AI-based seagrass morphology measurement. In Journal of Environmental Management, pages 122246, 2024. **Q1 Ranked Journal**, (Impact Factor: 8.0). Publisher: Elsevier.

11 **Sajal Halder**, Md Rafiqul Islam, Quazi Mamun, Arash Mahboubi, Patrick Walsh, Md Zahidul Islam A Comprehensive Survey on AI-enabled Secure Social Industrial Internet of Things in the Agri-Food Supply Chain , In Journal of Smart Agricultural Technology, **Q1 Ranked Journal**, (Impact Factor: 6.3), Publisher: Elsevier.

12. **Sajal Halder**, Kwan Hui Lim, Jeffrey Chan and Xiuzhen Zhang. Transformer-based Multi-task Learning for Queuing Time Aware Next POI Recommendation. In the 25th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD-2021), **Core Rank: A**
13. **Sajal Halder**, Md. Samiullah and Young-Koo Lee. Supergraph based Periodic Patterns Mining in Dynamic Social Networks. Expert Systems with Applications-An International Journal. volume 72, pages 430-442, 2017, **Q1 Ranked Journal**, (Impact Factor - 6.954). Publisher: Elsevier.
14. MS Rahman, **Sajal Halder**, MA Uddin, UK Acharjee, An efficient hybrid system for anomaly detection in social networks. In Cybersecurity, **Q1 ranked Journal**, (Impact Factor - 6.27), 2021, Volume 4, Pages 1-11, Publisher: Springer.

15 M.A. Talukder, M. Khalid, M. Kazi, N.J. Muna, M. Nur-e Alam, **Sajal Halder**, N. Sultana, A hybrid cardiovascular arrhythmia disease detection using convnext-x models on electrocar-diogram signals. Scientific Reports, **Q1 Ranked Journal**, 2024.

Total Publications: 30, Citation: 840+, h-index: 17 and i10-index: 21 (Google Scholar Link)

PROFESSIONAL EXPERIENCE (IN AUSTRALIA)

Research Fellow, RMIT University
Melbourne, Australia — August 2025 – Current

- Lead a Defence-funded project focused on detecting and evaluating the authenticity of AI-generated content on social media.
- Build a framework to generate real life social media posts, bypassing LLM jailbreak using GPT, DeepSeek, Mistral LLM models for 10 campaigns.
- Utilize LLaMA-3, Gemma-3 and GPT-OSS-120B to assess content across five dimensions: consistency, influence, engagement, tone, and faithfulness.

- Built a multi-model pipeline integrating RoBERTa, DeBERTa, and DistilBERT with cloud-based data pipelines for real-time misinformation detection.
- Lead and mentor junior researchers while delivering actionable technical insights to Defence stakeholders through clear, professional communication.
- Achieved 87% detection accuracy on real-world campaign datasets and reduced latency by 30%, strengthening national security posture through near real-time threat assessment.

Skills: LLaMA 3.1, Jailbreak, Agentic LLMs, LangGraph, LangChain, Hugging Face, Generative AI, Model Fine-Tuning, Deep Learning, Python, Long-Context Reasoning, Data Visualization, Technical Writing, Teamwork.

Research Technician (Postdoctoral Fellow), Data61, CSIRO, Australia
Collaboration with Google February 2024 – August 2025

Working on a Google Australia–funded project with Data61, CSIRO (AUD $616K) to standardize and prioritize software vulnerabilities, aiming to reduce the attack surface and enhance the resilience of critical infrastructure across Australia.

- Working within Data61's cybersecurity research ecosystem, contributing to nationally significant projects addressing software supply-chain security and critical infrastructure resilience.
- Engaged in the development of advanced technology for detecting function-level vulnerabilities, leveraging innovative approaches.
- Actively curating a comprehensive vulnerability dataset by extracting insights from diverse ecosystems' schemas and commits.
- Innovated code chunk-based model for function-level vulnerability detection in C/C++ and Python.
- Created six datasets using patch info with LLMs and LLMs alone for vulnerability detection.
- Utilized fine tune model on code chunks to identify vulnerable packages.
- Proposed model achieved 94-96% accuracy and 93-96% F1 score across six datasets.
- **Achievement 1:** Sajal Halder, Muhammad Ejaz Ahmed, Seyit Camtepe, "FuncVul: An Effective Function Level Vulnerability Detection Model using LLM and Code Chunk". In the 30th European Symposium on Research in Computer Security (ESORICS) 2025, **Core Rank: A**.
- **Achievement 2:** Michael Robinson, Sajal Halder, Muhammad Ikram, Hyoungshick Kim, Seyit Camtepe, Muhammad Ejaz Ahmed, "Original Sin of npm: A Study on Vulnerability Propagation in JavaScript Dependency Networks". Accepted to 21st ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2026), **Core Rank: A**.
- **Achievement 3:** Sajal Halder, Muhammad Ejaz Ahmed, Seyit Camtepe, "FuncVul-BM: Function-Level Binary and Multi-Class Vulnerability Detection with LLM-Augmented GraphCodeBERT and CNN". Submitted to IEEE Transactions on Dependable and Secure Computing, **Q1 ranked** journal.

Skills: LLMOps, Hugging Face, Google AI Studio, Prompt Engineering, Generative AI, Gemini API, Model Fine Tuning OpenAI, CodeBERT, GraphCodeBERT, Deep Learning, Python, Report/Article Writing, Data Visualization and Teamwork.

Research Fellow, Charles Sturt University
Wagga Wagga, NSW, Australia. December 2022 – February 2024

Focused on the SOCRATES project from CSU, funded by the Cyber Security CRC with a total grant of AUD $2.12M, aimed to develop intelligent technologies and advanced methods to evaluate and enhance software security for critical systems.

- We developed a metadata-based malicious and benign package detection model for the NPM repository.
- We introduced two sets of features, easy to manipulate (ETM) and difficult to manipulate (DTM), where manipulating DTM depends on long-term planning and monotonic properties. Then, we verify our feature selection effectiveness based on four well-known machine learning and one deep learning technique.
- We also analysed algorithms' performance using metadata manipulation and recommend better metadata adversarial attack-resistant algorithms.
- Experiment analysis shows that our proposed model reduces 97.56% False Positive and 80.35 % False Negative number.
- **Achievement 1:** Sajal Halder et al."Malicious Package Detection using Metadata Information," In The Web Conference 2024 (**Core Rank: A***).
- **Achievement 2:** Sajal Halder et al."A comprehensive survey on AI-enabled secure social industrial Internet of Things in the agri-food supply chain," In Smart Agricultural Technology, 2025, **Q1 ranked** journal.

Skills: Feature Engineering, Data Scraping, Open Source Software, Machine Learning Model, Deep Learning, Python, Report/Article Writing, Data Visualization and Teamwork.

Research Assistant, Acciona & RMIT University
Melbourne, Australia. August 2022 – December 2022

- We developed an efficient attrition evaluation model the HR of one particular project comes to know about the employee who is willing to leave the company, then he/she can manage the number of hiring and can get the valuable asset whenever they need so for the efficient flow of work.
- Implemented a data pipeline architecture to efficiently collect and transform employee data into a machine-readable format. The process encompassed various stages, including data cleaning, feature engineering, handling missing data, conducting exploratory data analysis, and studying correlations.
- Designed a comprehensive attrition cost model that assesses the financial and non-financial impact of employee turnover, accounting for tangible and intangible expenses incurred in the replacement process.

Skills: Machine Learning Model, Deep Learning, Python and R Programming, Report/Article Writing, Presentations, Data Visualization and Teamwork

Research Worker (Casual), CQ University
Melbourne, Australia. May 2022 – June 2023

- We constructed an annotated dataset of seagrass objects, employing the image annotation platform Roboflow, renowned for its efficiency.
- We devised a seagrass object classification model using advanced deep learning techniques, effectively detecting the three distinct object categories.
- Furthermore, we introduced a dynamic model tailored for quantifying object morphology, encompassing dimensions such as height and width.

- Analysis of our experimental results underscores the efficacy of our model, with an average recall of 97.5%, average precision of 83.3%, and an average F1-score of 90.1%.
- **Achievement 1:** Sajal Halder et al."AI-based seagrass morphology measurement," In Journal of Environmental Management, 2024, **Q1-ranked** journal.

Skills: YOLO6 Object Detection, Roboflow, Python, Deep Learning and Machine Learning

Tutor, RMIT University, Australia. March 2019 - October 2020

Dept. of Computer Science, School of Science

PhD Researcher, RMIT University
Melbourne, Australia. August 2018 – August 2022

i. Project Title: Capacity aware Fair POI Recommendation using Users Satisfaction.
Innovation: Design a novel fair POI recommendation model considering POI capacity and users preferences.
Language and Tools: Python, Actor-critic network, GCN, Tensorflow, Neural Network, Graph Allocation and Optimization
Achievement 1: Sajal Halder, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. Capacity-aware fair POI recommendation combining transformer neural networks and resource allocation policy. In Applied Soft Computing Journal, pages 110720, 2023, **Q1 Ranked Journal**, (Impact Factor – 8.7). Publisher: Elsevier.

ii. Project Title: Deep reinforcement learning of dynamic POI generation and optimization for itinerary recommendation
Innovation: Propose and implement dynamic personalized POIs and constructs itineraries within the budget time maximising user interest and minimising queuing time named DLIR.
Language and Tools: Python, AWS, GCN, Tensorflow, Neural Network
Achievement 1: Sajal Halder, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang.Deep learning of dynamic POI generation and optimisation for itinerary recommendation. In **ACM Transaction on Recommendation Systems**, 2025.

iii. Project Title: POI Recommendation with Queuing Time and User Interest Awareness
Innovation: Improve users interest based multi-tasking based transformer architecture models (TLR_U and TLR-M_UI) for best POI recommendation.
Language and Tools: Python, PyCharm, Jupyter, Tensorflow, Keras, Matplotlib, Transformer, Lucid Chart, Overleaf
Achievement 1: Sajal Halder, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. POI Recommendation with Queuing Time and User Interest Awareness. In Data Mining and Knowledge Discovery, 2022, **Q1 Ranked Journal**, (Impact Factor - 5.406). Publisher: Springer.

iv. Project Title: Transformer-based Multi-tasking for Queuing Time Aware Next POI Recommendation
Innovation: Design and implement multi-tasking based transformer architecture models (TLR and TLR-M) for better POI recommendation.
Language and Tools: Python, PyCharm, Jupyter, Tensorflow, Keras, Matplotlib, Transformer, Lucid Chart, Overleaf
Achievement 1: Sajal Halder, Kwan Hui Lim, Jeffrey Chan and Xiuzhen Zhang. Transformer-based Multi-task Learning for Queuing Time Aware Next POI Recommendation. In the 25th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD-2021), **Core Rank: A**

v. Project Title: Efficient Itinerary Recommendation via Personalized POI Selection and Pruning
Innovation: Develop efficient personalized itinerary recommendation model (EffiTourRec).
Language and Tools: R-studio, Lucid Chart, Overleaf, MatLab.
Achievement 1: Sajal Halder, Kwan Hui Lim, Jeffrey Chan, and Xiuzhen Zhang. Efficient Itinerary

Recommendation via Personalised POI Selection and Pruning. In Knowledge and Information Systems, Volume 64, Pages 963–993, 2022. **Q1 Ranked Journal**, (Impact Factor - 2.53), Publisher: Springer.

PROFESSIONAL EXPERIENCE (IN ABROAD)

Lecturer/Assistant Professor July 2016 - July 2018
Dept. of Computer Science and Engineering,
Jagannath University, Dhaka-1100, Bangladesh.

Lecturer/ Assistant Professor October 2013 - July 2016
Dept. of Computer Science and Engineering,
Bangabandhu Sheikh Mujibur Rahman Science & Technology University (BSMRSTU),
Gopalganj, Bangladesh.

Software Developer August 2012 - February 2013
Neighbor System Ltd, Seoul, South Korea.

PhD CO-SUPERVISOR EXPERIENCE

Student 1: Sishun Liu (PhD, 4th Year), RMIT University, Australia December 2025 - Current
Student 1: Mounika Badulla (PhD, 3rd Year), CQ University, Australia June 2024 - Current
Student 2: Zeeshan Bin Siddique (PhD, 1st Year), CSU, Australia January 2025 - Current

TAUGHT COURSES

Course Tutor / Lab Instructor (In Australia): Algorithms and Analysis (Semester 1, 2019 and Semester 2, 2019), Software Engineering (Semester 2, 2019)
Course Teacher (in Bangladesh): Computer Fundamentals, Discrete Mathematics, Programming with C / C++, Digital System, Design and Analysis of Algorithms, Data Mining and Warehouse, Machine Learning, Big Data Analytics, Advanced Data Mining and Machine Learning

EDUCATION

Doctor of Philosophy (PhD) Date of Graduation: December 2022
School of Computing Technologies,
RMIT University, Melbourne, Australia.
Thesis: Itinerary Recommendation based on Deep Learning.

Master of Engineering Date of Graduation: August 2013
Computer Engineering,
Kyung Hee University, South Korea
Thesis : Supergraph based Periodic Behaviors Mining in Dynamic Social Networks.

Bachelor of Science Date of Graduation: November 2010
Computer Science and Engineering
University of Dhaka, Bangladesh

SCHOLARSHIPS

- RMIT International Postgraduate Research Scholarship (August 2018 - April 2022)
- RMIT Research Stipend (RRSS-SC) (August 2018 - April 2022)
- Korean Global IT Scholarship (September 2011 - August 2013).
- Kyung Hee University President Scholarship (September 2011 - August 2013).

JOURNAL/ CONFERENCE REVIEWERS/PC MEMBER

- European Conference on Artificial Intelligence 2024, 2025 (Rank: A), PC Member
- Engineering Applications of Artificial Intelligence (Q1, IF: 7.8)
- Information Process and Management (Q1, IF: 7.47) | Applied Soft Computing (Q1, IF: 8.7)
- International Journal of Machine Learning and Cybernetics (Q1, IF:5.6)
- Scientific Reports (Q1, IF: 4.6) | IEEE Access (Q1, IF 3.367)
- IEEE Transactions on Computational Social Systems (Q1, IF: 4.74)
- Transactions on Knowledge Discovery from Data (Q1, IF: 4.2)
- Expert System with Applications, (Q1, IF 6.954) | World Wide Web (Q1, IF: 3.00)
- Knowledge and Information Systems (Q1, IF: 2.82)
- Neural Processing Letters (Q1, IF: 3.1) | Array (Q1, IF: 2.3)
- European Conference on Artificial Intelligence (ECAI) (Rank A, 2024-2026)

Other Activities

- Assistant Provost, Bijoy Dibosh Hall, BSMRSTU, Bangladesh (March 2015- July 2016).
- Deputy Sub-Project Manager (DSPM), Establishment Campus Network on BSMRSTU, Bangladesh, HEQEP Project. (Jun 2015-Jul 2016).
- Member, Establishment Campus Network on Jagannath University, Bangladesh, HEQEP Project. (Aug 2016- Jul 2018).
- Treasurer, CSE Association, Bangabandhu Sheikh Mujibur Rahman Science & Technology University (Jan 2015- Jul 2016).